Dhruv Tiwari

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Education

Sardar Vallabhbhai National Institute of Technology, Surat **Dec 2021 – July 2026**

Integrated M.Sc. Physics — ***CGPA: 9.06/10.0***

Relevant Coursework: Machine Learning in Physics (Python), Fundamentals of Computer Programming (C/C++), MATLAB Programming

Technical Skills

Programming Languages: Python, C++
ML/AI Frameworks: TensorFlow, PyTorch, Langchain, HuggingFace, OpenAI SDK, CrewAI
Data Science Tools: Pandas, NumPy, Matplotlib, Seaborn, Tableau
MLOps & Development: Vercel, AWS, Git/GitHub, MLflow, DagsHub
Databases: SQL, SQLite, Pinecone
Core Competencies: Feature Engineering, Model Training, Generative AI, Agentic AI

Experience

Master's Thesis Researcher **Jan 2026 – June 2026**

SVNIT, Surat

- Designed an ML pipeline to predict low-frequency electrical impedance of Perovskite materials from high-frequency data alone, tackling 242 highly correlated, multi-scale target variables
- Built a custom PyTorch Wide ResNet architecture using Target PCA to compress 242 outputs into 10 components and a signed log transformation for extreme value ranges, achieving 94% Log-Space R^2 while cutting training time from 45 to under 9 minutes
- Replaces slow, hours-long lab measurements with instant predictions, accelerating development of next-gen solar cells and optoelectronic devices

Projects

Local RAG Pipeline for Document Q&A | *Ollama, HuggingFace, Langchain, Pinecone, FAISS*

- Architected a complete Retrieval-Augmented Generation (RAG) system enabling accurate context-specific question answering from PDF documents with 100% data privacy
- Implemented vector embeddings and semantic search using open-source LLM (Llama3.1) and HuggingFace models for efficient document retrieval and response generation
- Validated system performance on Facebook's RAG research paper, extracting relevant context and generating precise answers to complex technical queries

Network Security Phishing Prediction System | *Python, Scikit-Learn, Scipy, Dagshub, FastAPI, AWS S3*

- Developed a modular prediction pipeline to detect malicious URLs using network features (SSL state, IP Address, etc.), integrating automated data validation and transformation components
- Conducted model training with multiple classification algorithms and systematically tracked performance using MLflow and DagsHub
- Achieved best test performance of 97% F1 score (XGBoost) and deployed the final predictive model and artifacts to an AWS S3 bucket

Co-Curricular Activities

Google Developer Student Club (GDSC), SVNIT **2022 – 2023**

Core Team Member *Surat, Gujarat*

- Secured partnership with 2 sponsors for Google's Student Development Club and successfully organized Hack The Tank 2023, the flagship hackathon with 100+ participants
- Coordinated 2 technical workshops on cloud computing and machine learning for student community engagement

Certifications

- Machine Learning Professional Certificate - Anaconda & LinkedIn
- Data Science, Generative AI Certificate - Udemy